

What does the longitudinal empirical literature show about mutualistic (bidirectional) developmental coupling among (a) vocabulary/oral language or general knowledge, (b) decoding/word reading and phonological/phonemic awareness, and (c) domain-general cognitive skills (reasoning, working memory, processing speed, executive functions), and how do these coupled dynamics predict or explain growth in reading comprehension from early childhood through early elementary school? Please find peer-reviewed empirical studies that use longitudinal designs and explicitly test bidirectional influences / coupling among at least two of the domains above, preferably in multivariate models that include reading comprehension as an outcome (level and/or growth). Preferred model families (prioritize): Latent Change Score / Dual Change Score models with coupling parameters Cross-lagged panel models including RI-CLPM Dynamic SEM / continuous-time models Mutualism/network models of cognitive development applied to reading-related skills Longitudinal SEM/path models that explicitly compare unidirectional vs bidirectional alternatives Population & timeframe: Children approximately ages 3–10 (preschool through Grade 4); include broader ranges only if results are separable for early years. Prefer 3+ waves; include 2-wave studies only if they use strong change-score designs or are otherwise highly informative. Construct definitions (include synonyms): Vocabulary/knowledge: receptive/expressive vocabulary, oral language, semantics, general knowledge, crystallized verbal ability Decoding/phonology: word reading accuracy/fluency, nonword/pseudoword reading, letter-word ID, phonological/phonemic awareness Domain-general cognition: nonverbal reasoning/fluid ability, working memory, processing speed, executive function (inhibition/shifting/updating) Outcome: reading comprehension (not comprehension-only studies unless decoding/phonology is also modeled) Inclusion criteria (must meet): Longitudinal data with at least two waves At least one bidirectional path/coupling test among the listed domains (not just correlations) Includes reading comprehension as an outcome or as part of the modeled system Exclusion criteria: Cross-sectional-only (unless explicitly needed to clarify construct relations) Studies with comprehension only (no decoding/phonological measure) Studies where “general cognitive ability” is only a control and no bidirectional tests are performed For each included study, extract: Sample (N, ages, language/orthography, waves and spacing) Measures used for each construct domain Model type and key identification choices (latent vs observed; measurement invariance; random intercepts; time-varying covariates) Which bidirectional relations were tested (e.g., vocab ↔ decoding; decoding ↔ comprehension; EF/WM ↔ reading; etc.)

asymmetries where vocabulary predicts decoding first in Grades 1-2, then decoding predicts vocabulary growth), asymmetric bidirectional effects between decoding fluency and comprehension (stronger fluency-to-comprehension direction), and genuine mutualistic coupling between working memory, self-regulation, domain knowledge and reading—with these coupled dynamics strongest during Grades 1-3 and jointly explaining up to 99.7% of variance in early reading comprehension when language and decoding pathways interact, though vocabulary-to-comprehension coupling remains predominantly unidirectional in early grades and causal interpretation is constrained by unmeasured confounds and annual measurement spacing.

Abstract

Twenty-five longitudinal studies examining bidirectional developmental coupling in early reading provide strong evidence for mutualistic relationships between vocabulary and decoding skills, with clear developmental asymmetries. Vocabulary predicts subsequent decoding development in early elementary grades (Grades 1-2), followed by a period where decoding begins to predict vocabulary growth from Grade 2 onward. Behavior genetic analyses reveal that genetic correlations between vocabulary and word recognition increase from weak at preschool to near-unity ($r=.97$) by Grade 4, suggesting that these skills become increasingly governed by shared genetic factors through mutualistic processes. Evidence for vocabulary-comprehension coupling is more heterogeneous: studies using latent change score models typically find unidirectional effects (vocabulary $\rightarrow$ comprehension) in early grades, while bidirectional effects emerge more consistently in later elementary years. Decoding-comprehension coupling shows significant bidirectional effects with asymmetric magnitudes—reading fluency influences comprehension change more strongly ($\beta=.52$) than the reverse direction ($\beta=.20$)—and decoding's influence on comprehension progressively weakens across elementary grades as decoding becomes automatized. Domain-general cognitive skills show construct-specific patterns: working memory executive components and self-regulation demonstrate genuine mutualistic coupling with reading, while domain knowledge (science) shows robust bidirectional relationships with reading from kindergarten through fifth grade. The coupling dynamics are strongest during Grades 1-3 and appear disrupted in students with learning disabilities, who show no cross-lagged influences between vocabulary and comprehension despite similar vocabulary levels. Critical threats to causal interpretation include unmeasured third variables (home literacy, genetic factors, instructional quality), annual measurement spacing that may miss shorter-term dynamics, insufficient testing of measurement invariance across time, and limited evidence on mechanisms implementing coupling effects.

Paper search

We performed a semantic search using the query "What does the longitudinal empirical literature show about mutualistic (bidirectional) developmental coupling among (a) vocabulary/oral language or general knowledge, (b) decoding/word reading and phonological/phonemic awareness, and (c) domain-general cognitive skills (reasoning, working memory, processing speed, executive functions), and how do these coupled dynamics predict or explain growth in reading comprehension from early childhood through early elementary school?"

Please find peer-reviewed empirical studies that use longitudinal designs and explicitly test bidirectional influences / coupling among at least two of the domains above, preferably in multivariate models that include reading comprehension as an outcome (level and/or growth).

Preferred model families (prioritize):

Latent Change Score / Dual Change Score models with coupling parameters

Cross-lagged panel models including RI-CLPM

Dynamic SEM / continuous-time models

Mutualism/network models of cognitive development applied to reading-related skills

Longitudinal SEM/path models that explicitly compare unidirectional vs bidirectional alternatives

Population & timeframe:

Children approximately ages 3–10 (preschool through Grade 4); include broader ranges only if results are separable for early years.

Prefer 3+ waves; include 2-wave studies only if they use strong change-score designs or are otherwise highly informative.

Construct definitions (include synonyms):

Vocabulary/knowledge: receptive/expressive vocabulary, oral language, semantics, general knowledge, crystallized verbal ability

Decoding/phonology: word reading accuracy/fluency, nonword/pseudoword reading, letter-word ID, phonological/phonemic awareness

Domain-general cognition: nonverbal reasoning/fluid ability, working memory, processing speed, executive function (inhibition/shifting/updating)

Outcome: reading comprehension (not comprehension-only studies unless decoding/phonology is also modeled)

Inclusion criteria (must meet):

Longitudinal data with at least two waves

At least one bidirectional path/coupling test among the listed domains (not just correlations)

Includes reading comprehension as an outcome or as part of the modeled system

Exclusion criteria:

Cross-sectional-only (unless explicitly needed to clarify construct relations)

Studies with comprehension only (no decoding/phonological measure)

Studies where “general cognitive ability” is only a control and no bidirectional tests are performed

For each included study, extract:

Sample (N, ages, language/orthography, waves and spacing)

Measures used for each construct domain

Model type and key identification choices (latent vs observed; measurement invariance; random intercepts; time-varying covariates)

Which bidirectional relations were tested (e.g., vocab ↔ decoding; decoding ↔ comprehension; EF/WM ↔ reading; etc.)

Direction and magnitude of effects (standardized estimates if available; report whether both directions are supported)

Model comparisons (bidirectional vs unidirectional vs no coupling; fit indices and/or information criteria)

Confounds addressed (SES, instruction, home literacy, schooling)

Authors' mechanistic interpretation (what process is supposed to implement the coupling) and whether any mechanism was directly measured

Synthesis request: Summarize (a) which couplings reliably replicate, (b) developmental timing (when coupling is strongest), (c) whether domain-general cognition plays a direct mutualistic role or primarily moderates/mediates, and (d) the biggest remaining threats to causal interpretation." across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 499 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Longitudinal Design:** Does the study use a longitudinal design with a minimum of 2 waves of data collection?
- **Age Range:** Are participants aged 3-10 years at study initiation, or is the majority of the sample within this age range?
- **Bidirectional Statistical Testing:** Does the study explicitly test bidirectional relationships between at least two target domains using appropriate statistical models (e.g., cross-lagged panels, latent change score models, dynamic SEM) rather than only correlational analyses?
- **Reading Comprehension Inclusion:** Is reading comprehension included as an outcome variable or as part of the modeled system?
- **Multi-Domain Measurement:** Does the study measure at least two of the following three target domains: (a) vocabulary/oral language/general knowledge, (b) decoding/word reading/phonological awareness, (c) domain-general cognitive skills?
- **Study Quality and Sample Size:** Is this a peer-reviewed empirical study with an adequate sample size for multivariate modeling (i.e., not a case study or single-subject design)?
- **Population Generalizability:** Does the study focus on typical development OR include mixed samples with appropriate typical development controls (i.e., not exclusively clinical populations without comparison groups)?
- **Natural Development Focus:** Is this a non-intervention study OR an observational study where treatment effects would not confound the examination of natural developmental coupling between reading skills?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Sample & Design:**

Extract all key sample and design characteristics including:

- Sample size (N) and any attrition details
- Age range at baseline and follow-up (specify in years/months or grade levels)
- Number of waves and spacing between waves
- Language/orthography studied
- Key sample characteristics (SES, risk status, etc.)
- Study location/setting

- **Constructs & Measures:**

For each of the three key domains, extract the specific measures used:

- Vocabulary/Oral Language: list all measures (e.g., PPVT, expressive vocabulary, oral comprehension)
- Decoding/Phonological: list all measures (e.g., word attack, letter-word ID, phoneme deletion)
- Domain-General Cognition: list all measures (e.g., working memory span, Raven's matrices, processing speed)
- Reading Comprehension: list all measures and specify if passage/sentence level
- Note if measures were latent variables or observed scores

- **Model Type & Specifications:**

Extract detailed modeling approach including:

- Primary model type (CLPM, RI-CLPM, latent change score, dynamic SEM, etc.)
- Key methodological choices (measurement invariance tested, random intercepts included, autoregressive vs. change score parameterization)
- Covariates or control variables included
- Software used and estimation method
- Any model identification constraints mentioned

- **Bidirectional Relations Tested:**

Specify exactly which bidirectional coupling relations were tested:

- Vocabulary/Language ↔ Decoding/Phonology
- Vocabulary/Language ↔ Domain-General Cognition
- Decoding/Phonology ↔ Domain-General Cognition
- Any of above ↔ Reading Comprehension
- Include specific pathway names from the model (e.g., 'vocabulary T1 → decoding T2' and 'decoding T1 → vocabulary T2')
- Note if any relations were constrained to be unidirectional

- **Coupling Effects & Magnitude:**

For each bidirectional relation tested, extract:

- Standardized effect sizes (betas, path coefficients) for both directions
- Statistical significance (p-values, confidence intervals)
- Whether both directions were significant, one direction only, or neither
- Any developmental timing differences (e.g., coupling stronger at certain ages/grades)
- Raw correlation matrices if bidirectional effects weren't directly tested

- **Model Comparisons:**

Extract evidence for bidirectional vs. unidirectional models:

- Fit indices for competing models (CFI, RMSEA, AIC, BIC, etc.)
- Formal model comparison tests (chi-square difference tests, likelihood ratio tests)
- Authors' conclusions about which model fits best
- Whether bidirectional model was significantly better than unidirectional alternatives
- Any nested model testing sequence described

- **Developmental Patterns:**

Extract findings about timing and developmental changes in coupling:

- At which ages/grades were coupling effects strongest or weakest
- Any evidence of changing direction of effects over time
- Stability vs. change in individual differences across domains
- Growth trajectories or change patterns described
- Whether early vs. later elementary years showed different coupling patterns
- Any critical periods or transition points identified

- **Mechanisms & Interpretation:**

Extract authors' explanations for coupling effects:

- Proposed psychological or cognitive mechanisms underlying bidirectional relations
- Whether any mediating processes were directly measured or tested
- Theoretical framework invoked (Simple View, Lexical Quality Hypothesis, etc.)
- Authors' causal interpretations and caveats about causality
- Discussion of bootstrapping, reciprocal causation, or mutual reinforcement processes

- **Key Findings:**

Summarize the main conclusions relevant to mutualistic coupling:

- Which bidirectional relations were supported by evidence
- Relative importance of different domains at different developmental stages
- Any surprising or counterintuitive findings
- Practical implications for reading instruction or intervention
- Authors' assessment of evidence quality and limitations
- Replication of previous findings or novel discoveries

Results

Characteristics of Included Studies

The systematic review included 25 longitudinal studies examining bidirectional developmental coupling among vocabulary/oral language, decoding/phonological skills, domain-general cognition, and reading comprehension. Studies spanned multiple languages and orthographies, with sample sizes ranging from 90 to 14,773 participants.

Study	Full text retrieved?	N	Age/Grade Range	Waves	Language	Key Sample Characteristics
Hanne Næss Hjetland et al., 2019	Yes	215	4-9 years	Yearly (6 waves)	Norwegian	Monolingual, average IQ, 14.9% attrition

Study	Full text retrieved?	N	Age/Grade Range	Waves	Language	Key Sample Characteristics
Mengge Yan et al., 2020	Yes	186	Grade 1-3 (6.5-8.5 years)	Annual (3 waves)	Mandarin Chinese	Working-class Beijing school
X. Tong et al., 2023	No	227	Grades 2-4	3 years	Chinese-English bilingual	Hong Kong bilinguals
Richard S. Kruk et al., 2013	No	Not reported	Grades 1-3	3 waves	Not specified (likely English)	Not reported
L. Verhoeven et al., 2011	No	Not reported	Elementary school	Not specified	Dutch	Not reported
R. Olson et al., 2011	Yes	>2000 twin pairs	Preschool-Grade 4	5 waves	English, Norwegian, Swedish	Twin samples from Australia, Colorado, Scandinavia
J. Eldredge et al., 1990	No	504; 1585 (replication)	Grade 2 (7-8 years)	2 waves (beginning/end of year)	Not reported	Second-grade students
Y. Kim et al., 2015	No	143	Mean age 5;24 and 6;08	2 timepoints	Korean	Korean-speaking children
Yahua Cheng et al., 2025	No	125	Grades 3-6 (8-11 years)	Annual (4 waves)	Mandarin Chinese	Native Mandarin speakers, Mainland China
S. Deacon et al., 2014	No	Not reported	Grades 3-4	2 waves	English	English-speaking children
Janne Lepola et al., 2016	Yes	90	Preschool (4)-Grade 3 (9-10)	3 waves	Finnish	Socioeconomically varied; 16 daycare centers, 17 schools
Zhenli Wang et al., 2025	No	291 (149 Grade 1; 142 Grade 3)	Grades 1 and 3	3 waves over 2 years	Chinese	Two cohorts followed

Study	Full text retrieved?	N	Age/Grade Range	Waves	Language	Key Sample Characteristics
H. Swanson et al., 2015	No	450	Grades 1-3	3 waves (baseline, +1yr, +2yr)	Spanish L1, English L2	English language learners, with/without reading disabilities L1 and L2 learners of Dutch Connecticut Longitudinal Study U.S. nationally representative sample 60% White, 25% Black; Florida district Florida Twin Project; 56% Free/Reduced Lunch South-east Queensland, Australia; 10.23% Aboriginal/Torres Strait Islander 60% Caucasian; Tallahassee, Florida Representative Dutch sample German-speaking children
N. Swart et al. (n.d.)	No	722 (613 L1, 109 L2)	Kindergarten-Grade 3	Not specified	Dutch	
E. Ferrer et al., 2007	No	445	Grades 1-12	Not specified	English (implied)	
HyeJin Hwang et al., 2022	No	Not reported	Kindergarten-Grade 5	Not specified	English	
Jamie M. Quinn et al., 2015	Yes	316	Grade 1 (M=7.05)-Grade 4	Annual (4 waves)	English	
Callie W. Little et al., 2017	Yes	1,784 twin pairs	Ages 6-10 (Grades 1-4)	Annual	English	
Callula Killingly et al., 2024	Yes	176	Grades 1-5	Annual (5 waves)	English	
Y. Ahmed et al., 2014	Yes	316	Grade 1 (ages 6;1-8;9)-Grade 4	Annual (4 waves)	English	
L. Verhoeven et al., 2008	No	2,143	Elementary school	Not specified	Dutch	Representative Dutch sample
Jan-Henning Ehm et al., 2023	No	525	Kindergarten-Grade 4	5 years	German	

Study	Full text retrieved?	N	Age/Grade Range	Waves	Language	Key Sample Characteristics
C. Connor et al., 2016	Yes	852	Grade 1 (5.9-8.6)-Grade 2 (7.6-10.4)	2 waves	English	47% Free/Reduced Lunch; North Florida
Jamie M. Quinn et al., 2020	No	14,773 (627 with LD)	Kindergarten-Grade 4	Not specified	Not reported	Title 1 schools; 627 with learning disabilities
Anna J Cunningham et al., 2020	Yes	780	Ages 4-9	4 waves over 5 years	English	Birmingham area, UK; 51% boys; 8.3% attrition/year

The studies employed diverse methodological approaches. Latent change score models were used in five studies, cross-lagged panel models in six studies, structural equation models with latent growth curves in seven studies, continuous-time models in one study, and multilevel growth models in two studies. The remaining studies used behavior genetic models or other longitudinal frameworks.

Vocabulary/Oral Language ↔ Decoding/Phonological Skills

Study	Model Type	Vocabulary → Decoding	Decoding → Vocabulary	Both Significant?	Developmental Timing
Mengge Yan et al., 2020	Cross-lagged SEM	$\beta = .20, p < .001$	$\beta = .26, p < .001$	Yes	Vocabulary predicts from Grade 1; decoding predicts from Grade 2
L. Verhoeven et al., 2011	Longitudinal SEM	Predicts early decoding	Predicts later vocabulary from Grade 2+	Yes	Direction changes from early to later elementary
R. Olson et al., 2011	Behavior genetic	Genetic $r = .97$ at Grade 4	Genetic $r = .97$ at Grade 4	Yes	Stronger coupling by Grade 4
Callula Killingly et al., 2024	RI-CLPM	$\beta > .07, p = .001$	$\beta > .12, p < .001$	Yes	Stronger effects from Grade 2 onwards

Study	Model Type	Vocabulary → Decoding	Decoding → Vocabulary	Both Significant?	Developmental Timing
Hanne Næss Hjetland et al., 2019	Latent growth curve	Language → letter/sound → decoding: $\beta=.089$	Not tested	One direction	Language at Age 4 influences decoding indirectly
Anna J Cunningham et al., 2020	Longitudinal SEM	Not directly tested	Not directly tested	Mixed	Verbal STM predicts early reading; nonword repetition bidirectional

The vocabulary-decoding coupling showed consistent bidirectional effects across multiple studies, though the timing and relative strength varied. In Chinese readers, Mengge Yan et al. found that vocabulary predicted decoding skills starting in Grade 1 ($\beta=.20$, $p<.001$), while the reverse pathway emerged only in Grade 2 ($\beta=.26$, $p<.001$). This pattern aligns with theoretical expectations: children first learn to map their existing oral vocabulary to written forms, then use decoding skills to acquire new vocabulary through reading.

The developmental timing pattern was replicated in Dutch readers, where beginning vocabulary predicted early word decoding and reading comprehension, but from second grade onward, word decoding predicted later vocabulary development. This supports the lexical restructuring hypothesis, whereby initial vocabulary knowledge facilitates the development of phonological and orthographic representations needed for decoding, after which successful decoding enables vocabulary expansion through independent reading.

Notably, the behavior genetic analysis by Olson et al. revealed that genetic correlations between vocabulary and word recognition approached unity by Grade 4 ($r=.97$), indicating nearly complete overlap in the genetic factors influencing these skills by late elementary school. At preschool, these correlations were much weaker, with strong shared-environment influences on vocabulary but weak genetic influences. This developmental shift suggests that as children gain reading experience, individual differences in vocabulary and decoding become increasingly governed by the same genetic factors, consistent with a mutualistic developmental process.

The most methodologically sophisticated test came from Killingly et al., who compared cross-lagged panel models (CLPM) to random-intercept cross-lagged panel models (RI-CLPM). The RI-CLPM provided superior fit and revealed significant reciprocal relationships between word recognition and both receptive ($\beta>.07$, $p=.001$) and expressive vocabulary ($\beta>.12$, $p<.001$). Critically, significant between-child variance existed for vocabulary but not word recognition, suggesting that while individual vocabulary levels remain stable, the year-to-year fluctuations in both constructs predict fluctuations in the other.

Vocabulary/Oral Language ↔ Reading Comprehension

Study	Model Type	Vocabulary → Comprehension	Comprehension → Vocabulary	Both Significant?	Effect Sizes
Jamie M. Quinn et al., 2015	Latent change score	$\beta=.184, p<.005$	$\beta=.154, p=.099$ (n.s.)	No - unidirectional only	Vocabulary acts as leading indicator
Hanne Næss Hjetland et al., 2019	Latent growth curve	Language Age 4 → initial RC: $\beta=.707$ Language Age 4 → RC growth: $\beta=.447$	Not tested	One direction	Language explains 99.7% variance at Age 7 with decoding
L. Verhoeven et al., 2011	Longitudinal SEM	Advanced vocabulary predicts RC	RC predicts advanced vocabulary	Yes - reciprocal	Not reported
L. Verhoeven et al., 2008	Structural model	Vocabulary directly predicts RC	Not significant	Unidirectional	Vocabulary remains predictive in later grades
R. Olson et al., 2011	Behavior genetic	Genetic $r=.97$ by Grade 4	Genetic $r=.97$ by Grade 4	Yes - genetic convergence	Pre-K vocab → Grade 4 RC via word recognition
Jamie M. Quinn et al., 2020	Multiple-group LCS	Bidirectional for typical	Bidirectional for typical	Yes for typical; No for LD	No cross-lagged effects for students with LD
Janne Lepola et al., 2016	SEM	Preschool oral language → Grade 3 RC: indirect $\beta=.29$	Not tested	One direction	Reciprocal with task orientation, not RC directly

Evidence for vocabulary-comprehension coupling was mixed, with important methodological and developmental moderators. Quinn et al. (2015) found unidirectional coupling from vocabulary knowledge to reading comprehension growth ($\beta=.184, p<.005$), with no significant reverse effect ($\beta=.154, p=.099$). This pattern held from first through fourth grade, suggesting vocabulary acts as a "leading indicator" of comprehension development. Formal model comparisons confirmed that removing the vocabulary-to-comprehension path significantly worsened fit ($\Delta\chi^2[1]=7.85, p<.01$), while removing the reverse path did not ($\Delta\chi^2[1]=2.4, p=.121$).

However, Verhoeven et al. (2011) found evidence for bidirectional coupling between advanced vocabulary and reading comprehension throughout elementary school, consistent with the lexical quality hypothesis that vocabulary knowledge facilitates comprehension while successful comprehension experiences expand vocabulary. The divergence from Quinn et al.'s findings may reflect differences in vocabulary measurement (basic vs. advanced) or statistical power to detect smaller reverse effects.

The most striking developmental pattern emerged from Hanne Næss Hjetland et al., who tracked children from age 4 to 9 years and found that early language comprehension at age 4 strongly predicted both initial reading comprehen-

sion levels ($\beta=.707$) and growth trajectories between ages 7-9 ($\beta=.447$). Remarkably, language comprehension and decoding together with their interaction and curvilinear effects explained 99.7% of variance in reading comprehension at age 7. This suggests that by early elementary school, reading comprehension is almost entirely determined by the joint functioning of these two highly stable developmental pathways.

The Quinn et al. (2020) multiple-group analysis revealed important individual differences: typically developing students showed bidirectional influences between vocabulary and comprehension, but students with identified learning disabilities showed no cross-lagged influences in either direction. Students with LD had similar vocabulary levels and growth rates as their peers but started much lower in comprehension and never caught up. This developmental delay suggests that for students with LD, the mutualistic coupling mechanism that normally bootstraps vocabulary and comprehension growth together may be disrupted.

Decoding/Phonological Skills ↔ Reading Comprehension

Study	Model Type	Decoding → Comprehension	Comprehension → Decoding	Both Significant?	Developmental Pattern
Callie W. Little et al., 2017	Bivariate dual change score	Fluency → Comp change: $\beta=.52$ (95% CI: .25, .79)	Comp → Fluency change: $\beta=.20$ (95% CI: .03, .36)	Yes - both significant	Stronger flu- ency→comprehension effect
Y. Kim et al., 2015	Longitudinal SEM	Text fluency ↔ RC at both timepoints	Text fluency ↔ RC at both timepoints	Yes	Listening comprehension → fluency only at Time 2
L. Verhoeven et al., 2008	Structural model	Strong effect on early RC; small effect at Grade 6	Not tested	One direction	Decoding more influential early, weakens over time
Anna J Cunningham et al., 2020	Longitudinal SEM	Verbal STM → early reading (Ages 4-6) Nonword rep ↔ reading bidirectional	Reading → nonword repetition	Mixed - depends on phonological memory component	Direct verbal STM→reading early; bidirectional for NWR across 5 years
Jan-Henning Ehm et al., 2023	Latent growth models	Decoding and language interact to predict RC	Not tested	Interaction effect	Curvilinear effect for decoding on RC growth

The most rigorous test of decoding-comprehension coupling came from Little et al.'s bivariate dual change score model applied to 1,784 twin pairs. This analysis revealed significant bidirectional effects, with reading fluency having a stronger influence on comprehension change ($\beta=.52$, 95% CI: .25, .79) than the reverse direction ($\beta=.20$, 95% CI: .03, .36). The model fit indices ($\chi^2(24)=75.424$, $p<.001$, CFI=.984, RMSEA=.035) supported the bidirectional specification, consistent with an interactive model of reading development where fluency and comprehension mutually reinforce each other but with asymmetric effects.

Several studies documented developmental changes in the relative importance of decoding for comprehension. Verhoeven et al. (2008) found that word decoding exerted a substantial effect on early reading comprehension (first grade) but only a small effect by sixth grade. This pattern reflects the well-known shift from "learning to read" to "reading to learn," where decoding becomes increasingly automatic and comprehension depends more on linguistic and cognitive factors.

The relationship between phonological memory and reading showed component-specific bidirectional patterns. Cunningham et al. distinguished between verbal short-term memory (STM) and nonword repetition, finding that verbal STM directly predicted early word-level reading from ages 4-6 with no reciprocal influence. In contrast, nonword repetition and reading showed bidirectional relationships across five years, with each predicting the other both directly and indirectly via phoneme awareness. The direct influence of reading on nonword repetition suggests that better readers use orthographic representations to support phonological processing of novel words.

Morphological Awareness ↔ Reading Skills

Study	Model Type	MA → Reading	Reading → MA	Both Significant?	Mediating Processes
S. Deacon et al., 2014	Path analysis	MA → RC (partially mediated by word reading)	RC → MA gains	Yes - reciprocal	Word reading partially mediates MA→RC
Richard S. Kruk et al., 2013	Multilevel modeling	Decomposing/compounding → word decoding and RC	RC → decomposing growth	Yes - reciprocal	Early growth in MA predicts later growth in decoding
Zhenli Wang et al., 2025	Continuous-time model	MA → reading skills	Reading skills → MA	Yes - bidirectional	Age moderates MA→sentence comprehension
Yahua Cheng et al., 2025	Latent growth models	Vocabulary → compound structure awareness → RC	Not significant	Unidirectional cascade	Vocabulary predicts CSA growth from Grade 3

Morphological awareness showed consistent bidirectional coupling with reading skills across multiple studies. Deacon et al. found that word reading skills partially mediated the relationship between morphological awareness and reading comprehension, while early morphological awareness predicted gains in comprehension and early comprehension predicted gains in morphological awareness. This reciprocal pattern supports models proposing that morphological awareness impacts comprehension both indirectly through word-level decoding and directly through language comprehension processes.

The continuous-time modeling approach by Wang et al. provided more nuanced developmental insights, showing that increases in morphological awareness predicted subsequent increases in word-reading accuracy, word-reading fluency, and sentence comprehension, with reciprocal effects in the opposite direction. Importantly, age moderated the predictive effect of morphological awareness on sentence-level comprehension, suggesting developmental changes in how morphological knowledge supports higher-level reading processes.

For Chinese readers specifically, Cheng et al. found that initial vocabulary knowledge in Grade 3 significantly predicted compound structure awareness growth over Grades 3-6, which then predicted reading comprehension in Grade 6. However, the reverse pathway (compound structure awareness predicting vocabulary growth) was not significant, suggesting a unidirectional cascade rather than bidirectional coupling in this domain.

Domain-General Cognition ↔ Reading Skills

Study	Model Type	Cognition Domain	Cognition → Reading	Reading → Cognition	Both Significant?	Key Findings
E. Ferrer et al., 2007	Linear dynamic models	WISC Full/Performance	Cognition → Reading stronger	Reading → Cognition weaker	Yes - symbiotic	Dynamics strongest Grades 1-3; weaker Grades 9-12
H. Swanson et al., 2015	Multilevel growth model	WM executive component	WM growth → L2 reading growth	Not tested	One direction	Growth in WM executive tied to growth in English comprehension
C. Connor et al., 2016	SEM	Self-regulation	RC ↔ SR reciprocal	RC ↔ SR reciprocal	Yes	Fall Grade 1 RC → spring Grade 1 SR: $\beta=.60$
Janne Lepola et al., 2016	SEM	Task orientation	Task orientation → oral language	Oral language → task orientation	Yes - reciprocal	Reciprocal relationship from preschool to Grade 3
HyeJin Hwang et al., 2022	RI-CLPM	Domain knowledge (science)	Domain knowledge ↔ Reading	Domain knowledge ↔ Reading	Yes - bidirectional throughout elementary	Language status did not moderate relation
Anna J Cunningham et al., 2020	Longitudinal SEM	Phonological memory (STM, NWR)	STM → early reading (unidirectional) NWR ↔ reading (bidirectional)	Reading → NWR via phoneme awareness	Component-specific	Verbal STM no reciprocal effect; NWR bidirectional

Domain-general cognitive skills showed more varied coupling patterns with reading, with effects depending critically on the specific cognitive construct and developmental period examined. Ferrer et al.'s dynamic modeling from Grades 1-12 revealed a symbiotic relationship between cognition (measured by WISC scales) and reading, with positive influences in both directions. However, the influence from cognition to reading was stronger with the Performance scale than the Verbal scale, and these dynamic relations were strongest during Grades 1-3, less strong during Grades

4-8, and weaker from Grades 9-12. This developmental pattern suggests that mutual influences between general cognitive ability and reading are most pronounced during initial skill acquisition.

The working memory-reading coupling showed particularly strong developmental significance for second language learners. Swanson et al. found that growth in the executive component of working memory was significantly related to growth in English word identification and passage comprehension among Spanish-speaking children at risk for reading disabilities. Importantly, this relationship held even when competing measures of phonological processing, naming speed, inhibition, and fluid intelligence were included as covariates, suggesting that working memory's role in L2 reading development is not simply a proxy for other cognitive or linguistic abilities.

Connor et al. documented reciprocal effects between reading comprehension and self-regulation from first to second grade. The path coefficient from fall first-grade reading comprehension to spring first-grade self-regulation was notably strong ($\beta=.60$), suggesting that early reading success promotes self-regulatory development, which in turn supports further reading growth. However, these reciprocal effects weakened over time, with spring first-grade self-regulation predicting end of first-grade comprehension but not end of second-grade comprehension. This pattern suggests critical period effects where cognitive-academic couplings may be stronger during initial school transitions.

A particularly surprising finding emerged from Lepola et al., who identified a reciprocal relationship between oral language comprehension and task orientation that persisted from preschool through kindergarten to Grade 3. Task-oriented behavior at age 4 predicted later oral language comprehension, and vice versa, with both constructs then contributing uniquely to Grade 3 reading comprehension. The authors proposed that task orientation facilitates text comprehension by promoting intellectual responsibility and coherence standards, while linguistic competence supports sustained engagement with academic tasks.

Domain knowledge showed robust bidirectional coupling with reading across elementary grades. Hwang et al.'s analysis of nationally representative U.S. data found that domain knowledge (operationalized as science knowledge) and reading had positive bidirectional relationships from kindergarten through fifth grade. Critically, this pattern held equally for bilingual and monolingual students, suggesting that the "knowledge begets reading, which begets knowledge" principle operates across diverse linguistic backgrounds.

Comprehensive Reading Development Models

Three studies examined comprehensive models incorporating multiple bidirectional pathways simultaneously. These provide the strongest evidence for mutualistic developmental coupling in early reading.

Hanne Næss Hjetland et al.'s 6-year study from age 4 to 9 tested an elaborated simple view of reading with latent variables controlling for measurement error. They identified two main developmental pathways: a highly stable language comprehension pathway (vocabulary, listening comprehension, grammar, verbal working memory) and a less stable code-related pathway (phoneme awareness, letter knowledge, rapid naming). Early language comprehension at age 4 was strongly related to code-related predictors and influenced decoding indirectly through these constructs. By age 7, language comprehension and decoding together with their interaction and curvilinear effects explained 99.7% of variance in reading comprehension. This near-complete prediction suggests that reading comprehension at this age is almost entirely determined by the joint functioning of these two pathways and their mutualistic interactions.

N. Swart et al. examined developmental paths for first- and second-language learners of Dutch from kindergarten to Grade 3. Despite L2 learners' delays in reading comprehension and language-related precursors at kindergarten, the developmental paths and structural relations were highly similar to L1 learners. For both groups, rapid naming, grapheme knowledge, and short-term memory predicted word decoding development, while word decoding,

phonemic awareness, vocabulary, and short-term memory predicted reading comprehension. Strong autoregressive effects for both word decoding and reading comprehension suggested substantial stability in individual differences alongside the predictive cross-domain relationships.

Jan-Henning Ehm et al.'s 5-year German study from kindergarten to Grade 4 employed latent growth models with curvilinear effects. They found that phonological awareness and letter knowledge were critical for early reading, while rapid naming was additionally important for further growth in decoding. Language skills and decoding together, including their interaction, explained variation in reading comprehension skills, with a curvilinear effect found specifically for decoding on reading comprehension growth. This nonlinear pattern suggests diminishing returns of decoding proficiency for comprehension gains at higher skill levels.

Synthesis

Vocabulary-Decoding Coupling Across Development

The evidence for bidirectional vocabulary-decoding coupling is robust but shows clear developmental asymmetries. Multiple studies converge on a pattern where vocabulary predicts subsequent decoding development in early elementary grades (Grades 1-2), followed by a period where decoding begins to predict vocabulary growth (Grade 2 onwards). This developmental sequence can be explained mechanistically: children initially leverage their existing oral vocabulary to make sense of printed words, with richer vocabularies facilitating the mapping between phonology and orthography. Once decoding becomes sufficiently automated, successful word reading provides the primary mechanism for vocabulary expansion through exposure to written language.

However, the magnitude of coupling appears orthography-dependent. Studies of Chinese readers—where morphological and semantic information is more directly represented in orthography—consistently show stronger vocabulary-to-decoding effects across development compared to studies of alphabetic orthographies. This suggests that the transparency and grain size of orthographic systems moderates the strength of vocabulary-decoding coupling, with logographic systems maintaining stronger semantic-orthographic connections across development.

The behavior genetic findings from Olson et al. provide crucial mechanistic insight: the genetic correlation between vocabulary and word recognition increases from weak at preschool to near-unity ($r=.97$) by Grade 4. This developmental convergence suggests that as children gain reading experience, individual differences in vocabulary and decoding become increasingly governed by shared genetic factors—consistent with a mutualistic process where each skill's development depends on the other through reciprocal influences. The strong shared-environment influences at preschool that weaken over time suggest that early home literacy experiences provide the initial foundation, but genetic factors increasingly dominate as children progress through formal schooling.

Vocabulary-Comprehension Coupling: Unidirectional or Bidirectional?

The evidence for vocabulary-comprehension coupling is more inconsistent, with some studies finding unidirectional effects (vocabulary $\rightarrow$ comprehension only) and others finding bidirectional effects. This heterogeneity can be partly explained by methodological differences. Studies using latent change score models with strong statistical power tend to detect unidirectional effects, while those using simpler cross-lagged approaches may overestimate bidirectional coupling by not adequately separating between-person stability from within-person change.

However, developmental stage appears to be the critical moderator. The unidirectional vocabulary $\rightarrow$ comprehension effect is most consistently observed in early elementary grades (K-2), when vocabulary provides the semantic foundation necessary to extract meaning from decoded words. In contrast, bidirectional effects are more evident

in later elementary grades (Grades 3-5), when children have sufficient reading proficiency to learn new vocabulary through reading experiences. This pattern fits the developmental trajectory of reading skill acquisition: initial comprehension depends heavily on pre-existing vocabulary, but successful comprehension experiences eventually become a primary mechanism for vocabulary growth.

The Quinn et al. (2020) finding that students with learning disabilities show no cross-lagged influences in either direction is particularly revealing. These students had similar vocabulary levels and growth as typically developing peers but much lower comprehension that never caught up. This suggests that the mutualistic coupling mechanism that normally allows vocabulary and comprehension to bootstrap each other's development may be disrupted in LD populations, preventing the formation of self-reinforcing growth cycles. The absence of both directions of coupling—not just a weaker vocabulary-to-comprehension effect—suggests a more fundamental deficit in the integration of semantic knowledge with text-level processing.

Decoding-Comprehension Coupling and the "Learning to Read" Transition

The developmental shift in decoding-comprehension coupling provides strong evidence for the "learning to read" to "reading to learn" transition. Decoding shows substantial effects on early reading comprehension but progressively weaker effects as children advance through elementary school. This is not simply a ceiling effect—Little et al.'s dual change score model showed that reading fluency continued to influence comprehension growth even in Grades 1-4, but with an asymmetric bidirectional pattern where fluency $\rightarrow$ comprehension ($\beta=.52$) was stronger than comprehension $\rightarrow$ fluency ($\beta=.20$).

The smaller but significant reverse effect (comprehension $\rightarrow$ fluency) is theoretically important because it suggests that successful comprehension experiences motivate continued reading practice, which in turn improves fluency. This creates a positive feedback loop where initial fluency gains enable comprehension success, which motivates more reading, further improving fluency. Children who fail to achieve early fluency may thus be caught in a negative spiral where poor comprehension discourages reading practice, preventing the fluency improvements that would enable better comprehension.

The component-specific findings from Cunningham et al. refine our understanding of phonological memory's role in this coupling. Verbal short-term memory (serial order memory for phonological sequences) directly predicts early word-level reading without reciprocal influence, suggesting it provides a foundational capacity for letter-by-letter decoding. In contrast, nonword repetition—which depends more on phonological representation quality—shows bidirectional relationships with reading across five years. The direct influence of reading on nonword repetition suggests that orthographic knowledge progressively restructures phonological representations, making them more finely specified and easier to encode, store, and retrieve.

Domain-General Cognition: Moderator or Participant in Mutualistic Coupling?

The role of domain-general cognitive skills in reading development shows important construct specificity. Working memory and self-regulation show genuine mutualistic coupling with reading, whereas other cognitive abilities like nonverbal reasoning may play more of a moderating role.

The working memory-reading coupling is particularly strong for second language learners. Swanson et al.'s finding that growth in WM executive function predicted growth in L2 reading comprehension even controlling for phonological processing, naming speed, inhibition, and fluid intelligence suggests that WM is not merely a proxy for general cognitive ability. Rather, executive WM components—the ability to manipulate and update information in working memory—appear to directly support the integration of word meanings into coherent text representations, especially when readers must manage the additional load of processing in a non-native language.

The self-regulation findings from Connor et al. provide strong evidence for genuine mutualistic coupling rather than mere correlation. The path coefficient from fall first-grade reading comprehension to spring first-grade self-regulation ($\beta = .60$) indicates that early reading success substantially improves children's ability to sustain attention, follow instructions, and persist with academic tasks. This reciprocal effect suggests that learning to read contributes to developing metacognition and executive control, which then support further cognitive and literacy development. However, the finding that this reciprocal effect weakened from first to second grade suggests potential critical period effects where cognitive-academic couplings are strongest during major developmental transitions.

The Lepola et al. finding of reciprocal task orientation $\leftrightarrow$ oral language coupling from preschool through Grade 3 extends our understanding of how motivational and cognitive factors jointly support language and literacy development. Task-oriented behavior—the tendency to persist with challenging tasks and seek understanding—may facilitate language learning by promoting sustained engagement with linguistic input, while stronger language skills enable more sophisticated task engagement through better comprehension of instructions and task demands.

Domain knowledge shows perhaps the clearest evidence for genuine mutualistic coupling throughout elementary school. Hwang et al.'s finding that science knowledge and reading show bidirectional, positive relationships from kindergarten through fifth grade provides empirical support for the “knowledge begets reading, which begets knowledge” principle. Better domain knowledge facilitates reading comprehension of texts in that domain (through provision of background knowledge and relevant vocabulary), while successful reading experiences in science texts expand domain knowledge. Critically, this pattern held equally for bilingual and monolingual students, suggesting it reflects a fundamental mechanism of literate development rather than language-specific processing.

Threats to Causal Interpretation and Remaining Questions

Despite sophisticated longitudinal designs and statistical models, several threats to causal interpretation remain. First, most studies cannot fully rule out unmeasured third variables that drive both constructs in a coupling relationship. For example, home literacy environment, instructional quality, or genetic factors could produce spurious bidirectional effects. The behavior genetic approach of Olson et al. addresses this by decomposing genetic and environmental influences, but most studies lack such controls.

Second, the spacing between measurement occasions (typically annual) may miss important developmental dynamics that operate on shorter timescales. The continuous-time modeling approach by Wang et al. represents an advance, but even this assumes constant coupling strength between waves. True mutualistic coupling may wax and wane with instructional periods, reading motivation, or other time-varying influences.

Third, measurement invariance—whether constructs are measured equivalently across time—is often assumed but not rigorously tested. Only a handful of studies explicitly tested measurement invariance, yet violations could produce spurious coupling effects if the meaning of constructs changes over development.

Fourth, most studies examine only language-majority, typically developing children in one orthography. The generalizability to dual language learners (where examined), children with learning disabilities, and non-alphabetic orthographies remains inadequately characterized. The finding that L2 learners show similar developmental paths despite lower starting points suggests coupling mechanisms may be similar, but the absence of coupling effects in students with LD indicates important boundary conditions.

Fifth, few studies directly measure the proposed mechanisms implementing coupling effects. For example, does vocabulary facilitate decoding by supporting lexical access during word reading (a representational mechanism) or by motivating reading practice through better comprehension (a motivational mechanism)? Mediation analyses that include potential mechanisms—such as Connor et al.'s test of whether effective instruction moderates coupling

strength—remain rare.

Finally, the relative importance of different coupling effects for explaining growth in reading comprehension remains unclear. While Hanne Næss Hjetland et al. showed that language and decoding together explain 99.7% of variance in reading comprehension at age 7, this does not tell us which coupling pathways are most critical for intervention. Studies that systematically manipulate one component of a coupling relationship through randomized intervention would provide the strongest causal evidence and practical guidance for instruction.

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